Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 8 AVSummary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26ME142A Unit 1 Class 8 AVSummary
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

UE26ME142A Unit 1 Class 8 AVSummary

1 UE26ME142A Unit 1 Class 8 AVSummary

Study focus: UE26ME142A_Unit 1_Class 8_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000166 · EV0000167

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Vengitesh from Civil Engineering and the Department of Science and Humanities, PS University.
- This is class number nine.
- In the last class, we had discussed about.
- The problems how we can solve under the topic moment.
- Today we will take up the next topic, which is couple.
- Two unlike parallel forces constitute a couple.
- When I say two, equal and unlike parallel forces constituble.
- When I say to equal, what does it mean?
- Means that the magnitude of the forces are same.
- That is what it means by equal.
- When I say unlike parallel, unlike parallel means.
- The sense of the two forces are opposite to each other, that is unlike.
- Then parallel, they are acting parallel to each other.
- One is moving into the screen, the other one is coming out of the screen.
- They are parallel.
- The magnitude of these two forces are the same.
- So a couple is defined as two equal, unlike parallel forces constitute a couple.
- This is one force which is the center of the forces coming like this.
- This is another force, the center of the force is going with this.
- But the magnitude of the force is F, the magnitude of this force is also F.

- The effect of a couple is to make the body rotate.
- No, how do you say that?
- No, since the sense of this force is like this, the sense of this force is like this, the combination of these two, when I sum it up, it will give me zero because the senses are zero, the sense are opposite.
- The magnitude of the force is the same, so effectively the net force acting is zero.
- But since they are acting at different.
- Points separated by some distance, it would make the body to rotate.
- The rotation can happen in the counterclockwise okay, or it can happen in the clockwise direction, depending upon how the sense of the forces are acting.
- if this forces to be acting along X and this to be acting along Y, then the Z will be coming out of the plane.
- The positive force means it is counterclockwise.
- It is counterclockwise.
- And vice versa, if it is going to rotate in the clockwise direction, it is anti-clockwise.
- It is not translatory motion, but it is rotatory motion.
- Whereas this force is acting at a distance of A plus D.
- The distance between the two forces is D.
- Whereas this force into this total distance A plus D would produce an anti-clockwise movement.
- So the movement produced at O because of this force is this F or it is R B into F in vector approach or R B cross F I should take, then it is plus RA cross.

Example or application from the lecture

- We will take this example and see what would be the movement that is going to be produced at point O.
- Let's take the perpendicular distance of this force from A to B R A, and perpendicular distance of this force from O to be R B.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 05:00–09:59

If I want to find out the movement of these two forces, the sense of this would be acting upwards.

Key details

- If I take the take these two forces to be acting in the horizontal plane, the vertical axis perpendicular to the horizontal plane.
- Will be the axis about which the movement is going to be produced, and you can see that this would produce a positive movement that is anti-clockwise movement.
- Also, I have explained.
- What is the effect of a couple?
- That is because the momentum produced by the couple at any point that you take would be the same.

- What is the effect of a couple?
- Then what is the classification of couple?
- And how do we determine the moment produced by a couple at any point, which should be equal to the.
- Distance between the two forces that is acting or constituting a couple into the magnitude of the force that is acting.
- There is a force F, negative force F acting at this point, that means the sense are different.
- They are differentiated or the distance between the two is D.
- The same force.
- F minus F and F, instead of applying it here, this is the point of application of these two.
- Whereas if I change it at this point, it would still be equal to M only.
- Change the position to be here and here, provided this distance is still D, again I will get the moment to be equal to M itself, and that is positive.
- You can see here this distance is half of this distance.
- So this is what is equivalent couple means.
- I already explained what is the characteristics, one of the characteristics.
- The net force acting because of a couple is zero.
- Because one force is acting in the positive direction, the other force is acting in the negative direction.
- So combination of these two will give me zero as the magnitude of the couple, or the force acting because of a couple is zero.
- The movement produced by a couple at any point is same.
- I have explained this also.
- The moment produced by a couple is same at any point.
- Or the movement produced by the couple at any point is same.
- Next, a couple.
- How do I balance a couple?
- I can balance a couple by an equal and opposite couple only.
- I cannot balance it by a single force because the net force acting is zero.
- A couple cannot be replaced by a single force.
- That is also a characteristics.
- These are the characteristics of a couple.
- What does this mean?
- The magnitude of the force is F, it is acting along this line of action, and the sense is moving away from A towards the right side.
- At B, what is the condition?
- The condition is that whatever effect is produced by this force acting at A, the same should be the effect produced by the system of force that's I am going to act at B.
- A that is the condition.
- That is what is called as a force couple system.

Example or application from the lecture

- So a couple is an example for a free vector.
- For this, I'm taking a simple example of a force acting at a point A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 10:00–14:59

So to do this, first thing what has to be done is I have to remove this force.

Key details

- To remove this force, what I'll do is I'll apply an equal and opposite force at B.
- The net effect of these two would be zero.
- But I cannot apply simply at P this force because it will change the.
- Effect of a force.
- I get only this again.
- But now I'll take this force minus F and F acting at A.
- Those two are now constituting a couple.
- I'll remove that and replace it by a couple, that is moment M, which is equal to F into D.
- We know that the moment produced by a couple is.
- Force into the distance between the two forces constituting a couple.
- The effect of this would be the same as the effect of this.
- Because this will have the same effect as this, because these two will cancel with each other.
- I will get only this.
- How I have replaced these two forces that is minus F and F.
- acting at A by a moment so this effect and this effect is same so eventually this effect and this effect are also same this is what is called as a force couple system and we will be using exclusively this concept for solving problems.
- I'll read out the statement of the problem.
- This is a solid problem in the textbook.
- The rigid structural member is subjected to a couple.
- Consisting of the 200 newton forces replace this couple by an equivalent couple constituting of the two forces P and P dash, each of which has a magnitude of 400 newton.
- Determine the proper angle T tar.
- If you refer to the diagram, you will be able to know.
- The point of application of the two forces of magnitude 100 newtons.
- One is positive, the other one is negative.
- P and minus P acting at these points at the top of the diagram.
- But the forces P is acting at an angle of theta with respect to this line.
- The effect of the forces 100 newtons, which is separated by 100 millimeter is going to produce a couple positive couple of magnitude 100 into 100.
- Because we have to replace that by this.
- along this line which will be equal to p cos theta.
- When I resolve this along the vertical.
- Which will be P sine theta, and this one will be P sin T again.
- They are acting in the same line and opposite to each other, so they will balance with each other.

- Whereas this and this, they are not acting in the same line, but they are separated by the distance 40 mm.
- Then get the value of P.
- From that equation I will be able to get the value of P.

Example or application from the lecture

- Next, let's take this simple example.
- They are separated by a distance of 100 millimeters because here it is given dimensions in millimeters.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next, let's take this example.

Key details

- By a force couple system acting at O.
- If I want to replace this force by force couple system at O, what is to be done?
- Just apply the 400 newton force, whatever is there acting at this point, bring it to this point, which will be 400 newtons acting at O like this, then determine.
- What is the movement produced by this force at O?
- How do we find that?
- We take this force, multiply it with the vertical distance between O and this point.
- Which would be equal to 200 sine of 60 degrees, which would be the vertical distance.
- The force couple system acting at O will be 400 newtons acting at O to the left, then a counterclockwise movement which will be equal to the magnitude of 400 into 200 sin 60.
- That is what is explained here.
- For this force, I will apply an equal and opposite force here.
- So that means simply bring this force here.
- Replace them by a moment.
- Which is equal to four hundred, that is this, into two hundred is this distance, sine sixty because this angle is sixty, the vertical distance is sine sixty.
- We have explained in this today's class, that is class number nine, what is a couple?
- What are the characteristics of a couple?
- What is the effect produced by a couple?
- Then how do we determine the moment produced by a couple?
- Then we have looked at what do we mean by force couple system.
- How do we get it?
- A force couple system.
- How do we solve problems using the force couple system?
- I hope all of you have understood.
- This is class number nine.
- We will continue to solve some more problems from the unsolved.

- Okay, in our next class that is class number 11.
- We will continue in the next class that is class number 11.
- My name is Dr.
- She Vingtej from Civil Engineering under Department of Science and Humanities, PS University.
- See you in the next class.

Example or application from the lecture

- The force coupled system is nothing but replacing a given force acting at a point A by some system acting at point B by a force couple system such that it produces the same effect, and we also have looked at.
- Okay, two problems, example problems I have taken and I have explained these two problems.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 8_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 8_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 8_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 8_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.

9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00039	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00040	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00041	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

C Revision and retrieval practice